Q:- $S = \{(1,1), (-1,2)\}$ (u_1, u_2) : Find $S^\perp$

A:- Let $v = (x,y)$ is orthogonal to u $\forall$ $u \in S$.

$\langle v, u_1 \rangle = 0$

$\langle v, u_2 \rangle = 0$

$\Rightarrow x + y = 0 \quad \text{---(i)}$

$-x + 2y = 0 \quad \text{---(i)}$

$x = -y$

$3y = 0$

$\Rightarrow y = 0$

$\therefore (x,y) = (0,0)$

$S^\perp = \{(0,0)\}$

Q:- $S = \{(1,1), (2,2)\}$

$x + y = 0$

$2x + 2y = 0$

$x = -y$

Let $x = \alpha$

$(x,y) = (\alpha, -\alpha) = \alpha(1,-1)$

$S^\perp = \{\alpha(1,-1), \alpha \in \mathbb{R}\}$

PKalika Maths